

# Data Communication

## Content 01: Introduction

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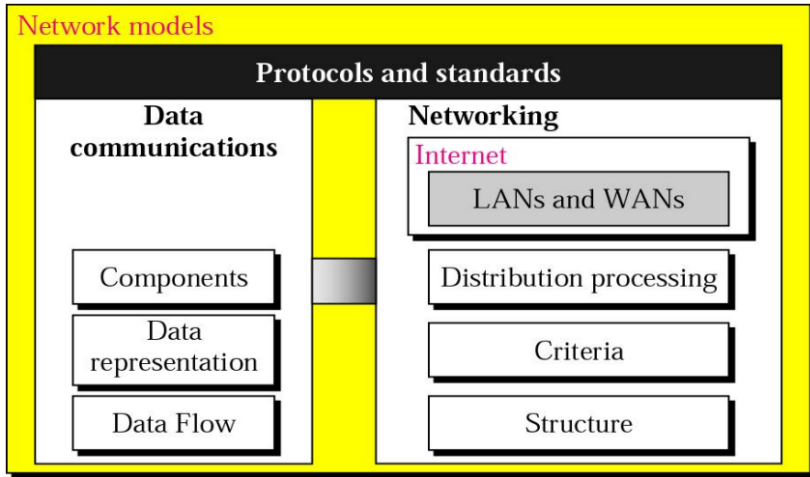
Department of Computer Science and Engineering

**United International University**



# Data Communication & Networking

## Network models



# Data Communications

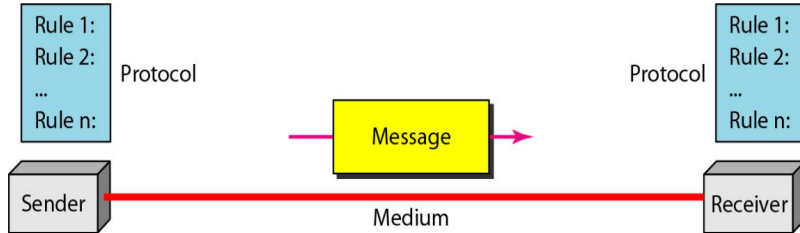
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- The term **telecommunication** means communication at a distance. The word **data** refers to information presented in whatever form is agreed upon by the parties creating and using the data.
- **Data communications** *are the exchange of data between two devices via some form of transmission medium such as a wire cable.*



# Components of Data Communication

There are **five** components of a successful data communication.



# Components of Data Communication

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- **Message:** is the information(data) to be communicated (text, numbers, pictures, sound, video or any combination of these).
- **Sender:** is the device that sends the data message (computer, workstation, telephone handset, video camera).
- **Receiver:** is the device that receives the message (computer, workstation, telephone handset, video camera).
- **Medium:** is the physically path by which a message travels from sender to receiver (twisted pair wire, coaxial cable, fiber- optic cable, laser , or radio waves(terrestrial or satellite microwave))
- **Protocol:** is a set of rules that govern data communication.



# Direction of Data Flow

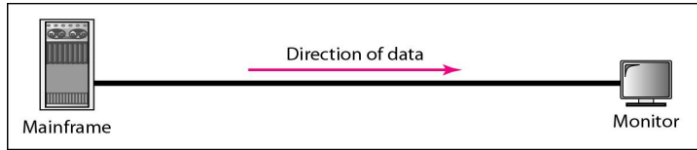
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Used to define the direction of signal flow between two link devices.

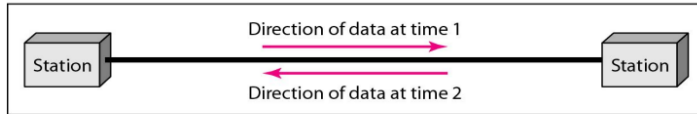
- **Simplex:** unidirectional, as on a one-way street (keyboard, monitor).
- **Half-Duplex:** each station can both transmit and receive, but not at the same time.
- **Full-Duplex:** both stations can transmit and receive simultaneously.



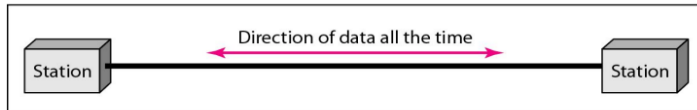
# Direction of Data Flow



a. Simplex



b. Half-duplex



c. Full-duplex

# Networks

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- A **network** is a set of devices (often referred to as nodes) connected by communication **links**.
- A **node** can be a computer, printer, or any other device capable of sending and/or receiving data generated by other nodes on the network.

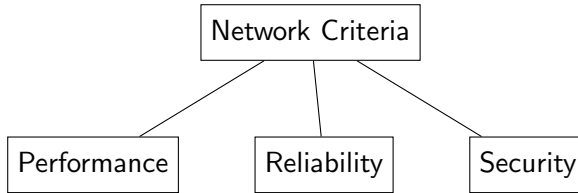




# Network Criteria

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Network criteria define the key factors that determine a network's effectiveness.



# Network Criteria

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## Performance

- Depends on a number of factors:
  - number of users
  - type of transmission medium(data rate
  - hardware
  - software : error-free delivery
- Evaluated by two networking metrics: **Throughput and Delay.**



# Network Criteria

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## Reliability

- Frequency of failure
- Recovery time of a network after a failure
- Catastrophe: Networks must be protected from catastrophic events such as fire, earthquake, or theft.

## Security

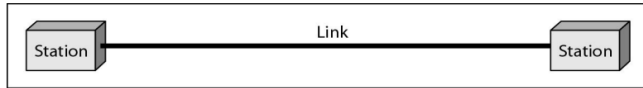
- Protecting data from unauthorized access
- Protecting data from Viruses and Worms



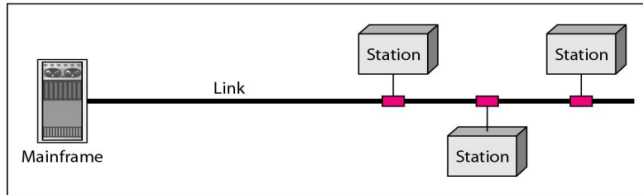
# Physical Structure

Types of Network Connection:

- **Point-to-point:** provides a dedicated link between two devices
- **Multipoint:** configuration in which more than two specific devices share a single link.



a. Point-to-point

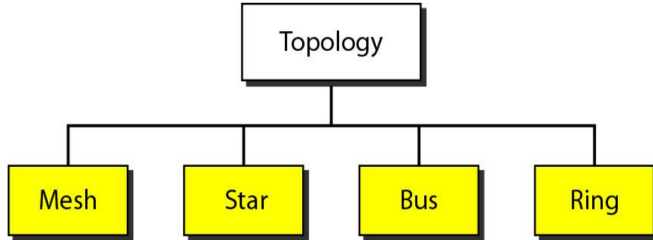


b. Multipoint

# Physical Topology

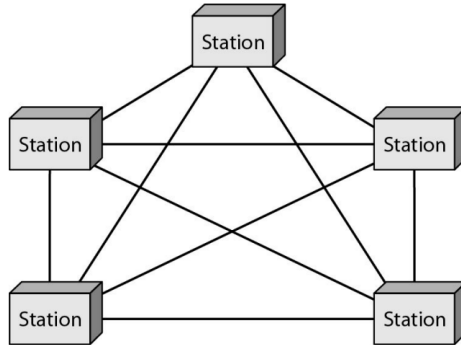
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Physical topology refers to the actual layout of devices and cables in a network. It defines how computers, switches, and other devices are physically connected.



# Mesh Topology

- Every device has a dedicated point-to-point link to every other device.
- A fully connected mesh network therefore has  $\frac{n(n-1)}{2}$  physical channels to link  $n$  devices.



# Mesh Topology

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## Advantages

- Dedicated links
- Robust
- Privacy and security
- Easy fault identification and fault isolation

## Disadvantages

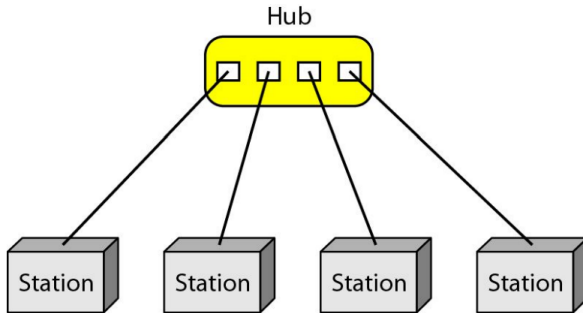
- Amount of cabling and the number of I/O ports
- Expensive



# Star Topology

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Each device has a dedicated point-to-point link only to a central controller, usually called a hub.





# Star Topology

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## Advantages

- Only one link and one I/O port to connect
- One link fails, only that link is affected

## Disadvantages

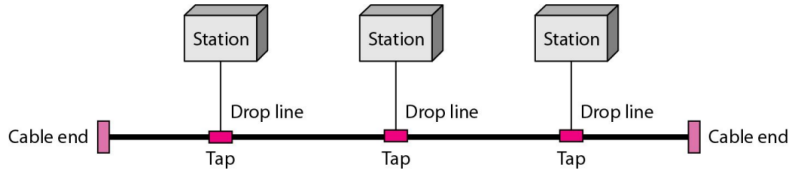
- Single Point of Failure
- Scalability Limitations



# Bus Topology

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One long cable acts as a backbone to link all the devices in the network.



# Bus Topology

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## Advantages

- Ease of installation

## Disadvantages

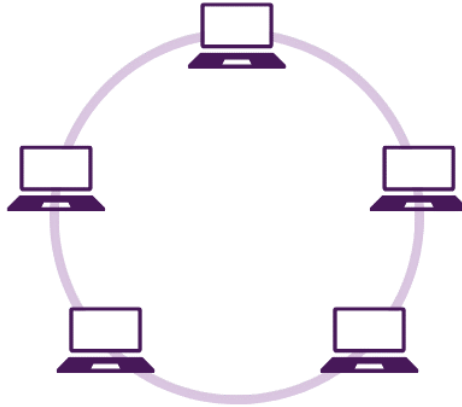
- Single Point of Failure
- Difficult reconfiguration and fault isolation



# Ring Topology

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Each device has a dedicated point-to-point line configuration only with the two devices on either side of it.



# Ring Topology

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## Advantages

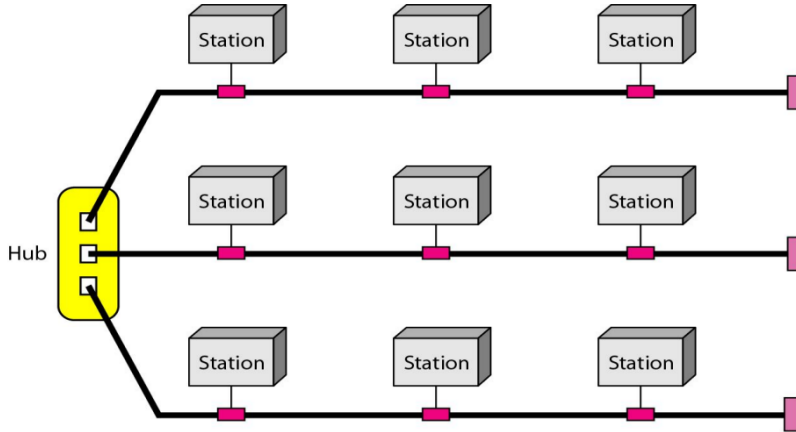
- Easy to install and reconfigure
- Simplified fault isolation

## Disadvantages

- Single Point of Failure
- Slow Data Transmission



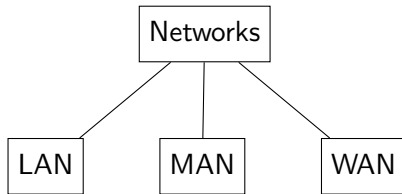
# Hybrid Topology



# Categories of Networks

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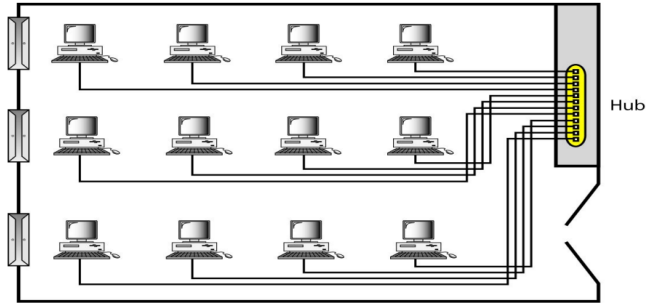
Based on the scope and geographical coverage of a network, it can be divided as following:



# LAN (Local Area Networks)

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LAN is usually privately owned and links the devices in a single office, building or campus.

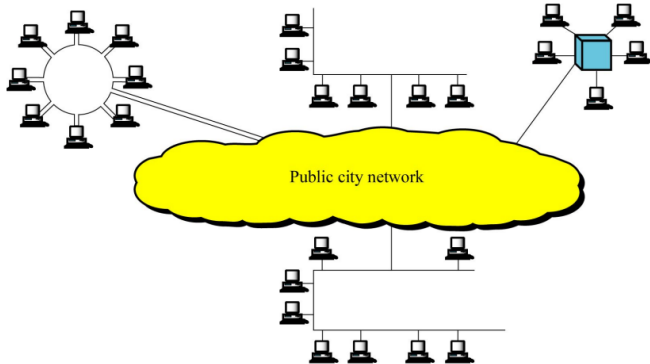




# MAN (Metropolitan Area Networks)

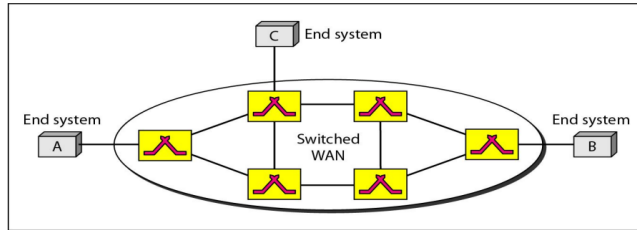
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MAN is designed to extend over an entire city.

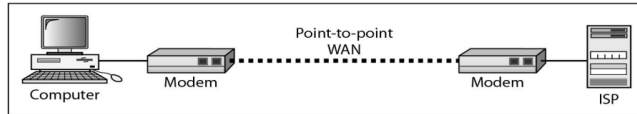


# WAN (Wide Area Networks)

WAN provides long-distance transmission of data over large geographical areas that may comprise a country, a continent, or even the whole world.



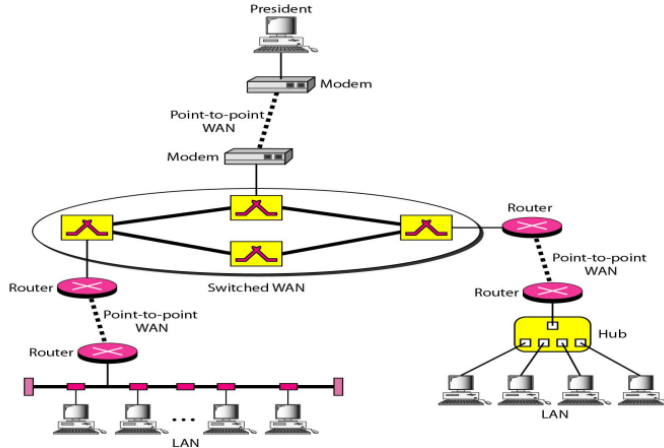
a. Switched WAN



b. Point-to-point WAN

# Interconnection of Networks: Internetwork

Two or more networks are connected, they become an internetwork.



# What is the Internet?

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- The Internet is a global internetwork that connects millions of computers worldwide.
- It enables communication, data sharing, and access to information through protocols like TCP/IP.
- Services like the World Wide Web, email, and cloud computing operate over the Internet.
- It consists of interconnected networks, including LANs, MANs, and WANs, forming a vast internetwork.



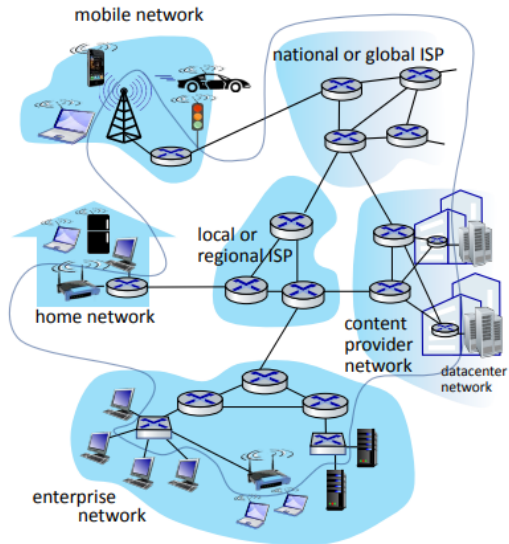
# Internet: A Brief History

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- **1960s:** The concept of interconnecting computers was proposed, leading to the development of early networking concepts.
- **1969:** ARPANET, funded by the U.S. Department of Defense, became the first operational packet-switching network.
- **1970s:** TCP/IP protocol suite was developed, providing a standard for data communication over networks.
- **1983:** The Internet Protocol (IP) and Transmission Control Protocol (TCP) were adopted as the standard, marking the birth of the modern Internet.
- **1990:** Tim Berners-Lee created the World Wide Web (WWW), making the Internet more accessible and user-friendly.
- **1990s:** Commercial use of the Internet expanded rapidly, and browsers like Netscape Navigator emerged.
- **2000s-Present:** The Internet evolved into a global platform for social media, e-commerce, streaming, and cloud computing.



# Internet Structure



# Protocols & Standards

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- **Protocols** are a set of rules that define how data is transmitted and received over a network, ensuring proper communication between devices. They are typically set by organizations like the Internet Engineering Task Force (IETF).
- **Standards** are widely accepted specifications that ensure interoperability and consistency across different devices, networks, and protocols. These are established by organizations such as the Institute of Electrical and Electronics Engineers (IEEE) and the International Telecommunication Union (ITU).



# Key Elements of a Protocol

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- **Syntax** refers to the structure or format of the data, meaning the order in which they are presented.
- **Semantics** refers to the meaning of each section of bits.
- **Timing** refers to two characteristics (when data should be sent and how fast it can be sent).





# Practice Problems

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- Data Communication and Networking (4th Ed) - Behrouz A. Forouzan  
Chapter 01 - 1, 3, 7, 9, 11, 12



# References

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- Data Communication and Networking (4th Ed) - Behrouz A. Forouzan  
Chapter 01 - 1.1, 1.2, 1.3, 1.4
- Computer Networking: A Top Down Approach (6th Ed) - Jim Kurose  
Chapter 01 - 1.1



# Thank You

